

WORK SHEET - 01

CHEMICAL BONDING

Single correct :

- Q.1 $2 \times \text{sulphuric acid} - \text{H}_2\text{O} + \text{O} \rightarrow \text{X}$
 $2 \times \text{sulphurous acid} - \text{H}_2\text{O} = \text{Y}$

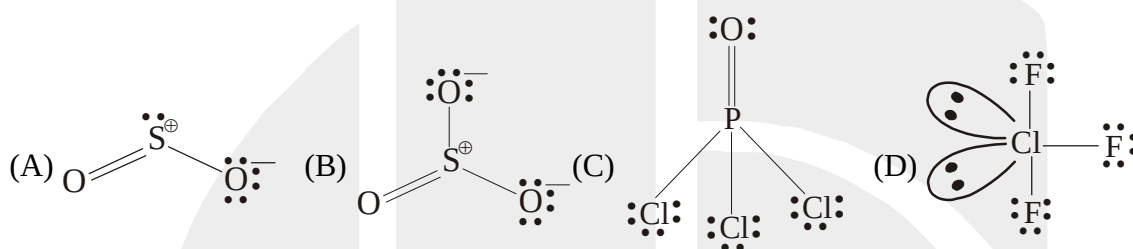
Which option is **correct** for above information?

- (A) X has S–O–S bond (B) Y has S–O–S bond
 (C) Y has S–S bond (D) X has S–S bond

- Q.2 Select the correct order with respect to bond angle.

- (A) $\text{NH}_3 < \text{NF}_3 < \text{NCl}_3$ (B) $\text{OH}_2 < \text{OF}_2 < \text{OCl}_2$
 (C) $\text{SH}_2 < \text{SF}_2 < \text{SCl}_2$ (D) $\text{BH}_3 < \text{BF}_3 < \text{BCl}_3$

- Q.3 Which is correct Lewis structure with formal charge on particular atom?



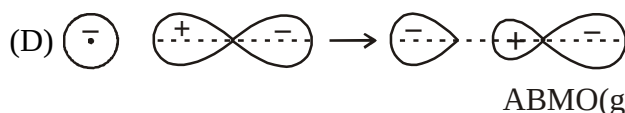
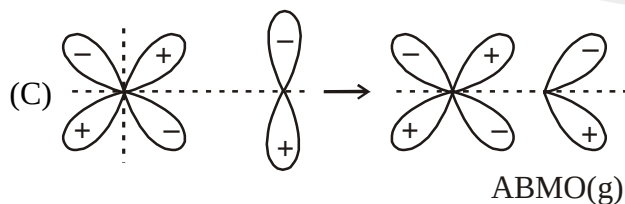
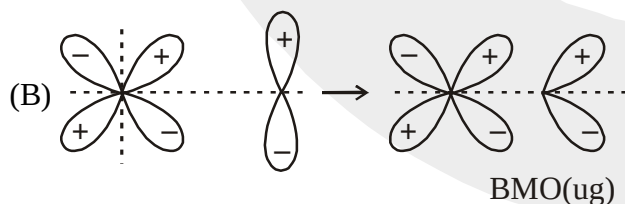
- Q.4 Which of the following statement is **correct** regarding $\text{C}_2\text{H}_2\text{F}_2$ molecule.

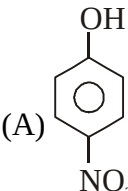
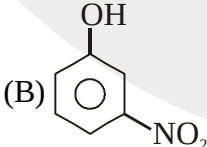
- (A) One isomer is polar, one is non polar (B) Two isomers are polar, one is non polar
 (C) Two isomers are planar, one is non planar (D) Two isomers are only possible and planar

- Q.5 Which of the following statement is **correct**.

- (A) O_2 is paramagnetic, O_3 is also paramagnetic
 (B) O_2 is paramagnetic, O_3 is diamagnetic
 (C) B_2 is paramagnetic, C_2 is also paramagnetic
 (D) Different observation is found in their bond length when $\text{NO} \rightarrow \text{NO}^+$ and $\text{CO} \rightarrow \text{CO}^+$

- Q.6 Which of the following is the correct combination of atomic orbitals according to MOT.



- Q.7 The bond length of the S–O bond is maximum in which of the following compound.
 SOBr_2 , SOCl_2 , SOF_2
 (A) SOCl_2 (B) SOBr_2 (C) SOF_2 (D) All have same length
- Q.8 Which one of the following molecules has zero dipole moment
 (A) ICl (B) AsCl_3 (C) Ni(CO)_4 (D) PCl_3
- Q.9 Select the molecule which has See-Saw shape.
 (A) XeOF_4 (B) $[\text{O}_2\text{IF}_2]^-$ (C) SOF_4 (D) POCl_3
- Q.10 Among KO_2 , KAlO_2 , CaO_2 and NO_2^+ , unpaired electron is present in:
 (A) NO_2^+ and CaO_2 (B) KO_2 and KAlO_2
 (C) KO_2 only (D) CaO_2 only
- Q.11 NH_3 & N_3H has hybridisation
 (A) sp^3 , sp (B) sp , sp^3 (C) sp^2 , sp (D) sp , sp
- Q.12 Give the correct order of initials **T** or **F** for following statements. Use **T** if statement is true and **F** if it is false.
 (I) Br–P–Br bond angle in PCl_3Br_2 is 180° .
 (II) NO_2^+ , NO_2 & NO_2^- species have O–N–O bond angle order, $\text{NO}_2^- < \text{NO}_2 < \text{NO}_2^+$.
 (III) Oxy acid is not produced, when NCl_3 undergoes hydrolysis.
 (IV) Silicones are very inert and water repellent.
 (A) TTTF (B) TFFT (C) FTFF (D) FTFT
- Q.13 The xenon compound(s) that are iso-structural with IBr_2^- and BrO_3^- respectively
 (A) linear XeF_2 and pyramidal XeO_3 (B) bent XeF_2 and pyramidal XeO_3
 (C) bent XeF_2 and planar XeO_3 (D) linear XeF_2 and tetrahedral XeO_3
- Q.14 How many molecules are linear in following compounds but does not have any lone pair on central atom.
 CO_2 , XeF_2 , $[\text{I(CN)}_2]^-$, $[\text{I}_2(\text{CN})]^+$, I_3^- , C_2H_2 , SnCl_2 , OF_2 , HgCl_2
 (A) 4 (B) 2 (C) 5 (D) 3
- Q.15 Which of the following has minimum boiling point?
 (A)  (B)  (C) CO_2 (D) SiO_2
- Q.16 Which of the following species is tetrahedral and coloured due to charge transfer spectrum?
 (A) VO_4^{3-} (B) CrO_2Cl_2 (C) $[\text{Sc(H}_2\text{O)}_6]^{3+}$ (D) both (A) and (B)
- Q.17 Choose the correct order of boiling point.
 (A) $\text{GeH}_4 > \text{SiH}_4 < \text{CH}_4$ (B) $\text{Xe} > \text{Ne} > \text{He}$
 (C) $\text{He} > \text{H}_2$ (D) $\text{CH}_4 > \text{CD}_4$

- Q.18 The **correct** order of dipole moment of HF, HCl, HBr, HI and H_2 is :
 (A) $H_2 < HF < HCl < HBr < HI$ (B) $H_2 < HF < HBr < HCl < HI$
 (C) $H_2 < HI < HBr < HCl < HF$ (D) $H_2 < HI < HBr < HF < HCl$

Assertion Reason

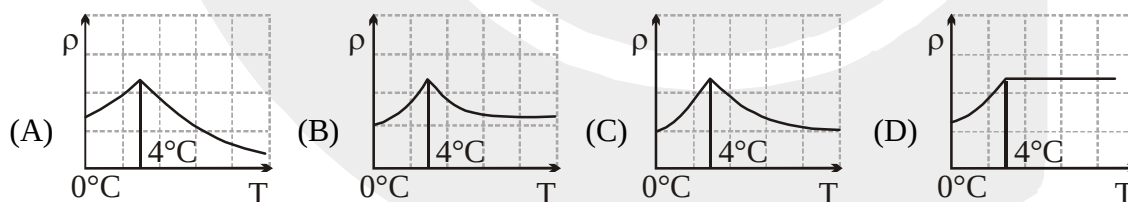
- Q.19 **Statement-1** : POF_3 exist but NOF_3 does not exist.
Statement-2 : 'P' can form five bond by expanding its octet while 'N' can not expand its octet to form five bonds.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.20 **Statement-1** : Red phosphorus is less volatile than white phosphorus.
Statement-2 : Red phosphorus has a discrete tetrahedral structure.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
- Q.21 **Statement-1** : Bond angle of HOF is lesser than HOCl.
Statement-2 : Oxygen atom is more electronegative than chlorine.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Comprehension

Paragraph for question nos. 22 & 23

Hydrogen bonding originates from the dipole-dipole interaction between H-atom and any of the other atom like F, O, N and in somecases with Cl atom. There are two types of H-bonding like intermolecular and intramolecular H-bonding.

- Q.22 Which of the following plot for ρ (density) v/s T (temperature) for liquid H_2O is **correct**:



- Q.23 Which of following statement is **incorrect**.
 (A) Boiling point of H_2O_2 is greater than that of H_2O
 (B) Ethylene glycol is less viscous than glycerol.
 (C) o-nitrophenol can be separated from its meta and para isomer using its steam volatile property.
 (D) In ice each 'O' atom is tetrahedrally arranged by four H-atoms which are all equidistant.

Paragraph for question nos. 24 to 26

The hybrid orbitals sp , sp^2 , sp^3 , sp^3d^2 , etc are equivalent, but there are some cases in which the hybrid orbitals are nonequivalent. This nonequivalence is found in sp^3d hybridisation. In sp^3d (trigonal bipyramidal) hybridisation, there are three equivalent hybrid orbitals developed from $(s + p_x + p_y)$ lie in the equatorial plane (i.e. xy -plane) at 120° , and other two equivalent hybrid orbitals developed from $(p_z + d_{z^2})$ project mutually at 180° in the axial directions. Thus there are two sets of hybrid orbitals, i.e. a set of three equivalent equatorial hybrid orbitals and a set of two equivalent axial hybrid orbitals. This idea has been established from the experimental findings in PCl_5 in which the three equatorial chlorines are different from the two axial chlorines. The equatorial P–Cl bond (204 pm) is shorter than the axial P–Cl bond (219 pm). If different atoms of different electronegativity are attached to the central atom, then **Bent's rule** will determine the positions of the different surrounding atoms in the trigonal bipyramidal geometry.

- Q.24 Which of the following statement is correct regarding the structure of $H_2CSF_2Me_2$ molecule.
(A) Two C–H bonds are in the same plane of equatorial S–Me bonds.
(B) Two C–H bonds are in the same plane of axial S–F bonds.
(C) Nodal plane of π -bond is perpendicular to the axial S–F plane.
(D) π -bond is formed by p_π – p_π overlapping.
- Q.25 In P_4O_{10} molecule, bridging P–O bond length is
(A) larger than that of in P_4O_6 (B) lesser than that of in P_4O_6
(C) Equal to that of in P_4O_6 (D) can not be compared
- Q.26 In which of the following pair of species, all bond angles are equal
(A) CO_3^{2-} , $COCl_2$ (B) PO_4^{3-} , $POCl_3$ (C) BF_4^- , BH_4^- (D) CH_3F , CH_4

Paragraph for question nos. 27 & 28

According to Fajan rules, the percentage of covalent character in an ionic compound increases if the cation is highly charged or the cation is small and the anion is large or the cation has pseudo inert gas configuration. As a result of increased covalent character, the melting point decreases and solubility in less polar solvent increases. Read and above comprehension and answer the following questions.

- Q.27 The correct order of increasing ionic character is
(A) $BeCl_2 < MgCl_2 < CaCl_2 < BaCl_2$ (B) $BeCl_2 < MgCl_2 < BaCl_2 < CaCl_2$
(C) $BeCl_2 < BaCl_2 < MgCl_2 < CaCl_2$ (D) $BaCl_2 < CaCl_2 < MgCl_2 < BeCl_2$
- Q.28 Based on lattice energy and other considerations, which one of the following alkali metal chlorides is expected to have the highest melting point ?
(A) LiCl (B) NaCl (C) KCl (D) RbCl

Paragraph for question nos. 29 to 31

When two compounds ACl_3 and DCl_3 of two elements A and D are mixed together a compound $ADCl_6$ is formed. Structural analysis showed that $ADCl_6$ is an ionic compound. Given that DCl_3 is trigonal planar and ACl_3 is trigonal pyramidal predict the shape of the anion.

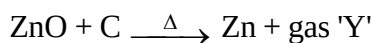
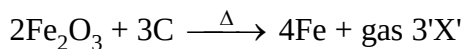
- Q.29 If anion has See - Saw shape then shape of cation formed is
(A) Linear (B) Bent
(C) Pentagonal bipyramidal (D) Trigonal planar
- Q.30 If shape of anion is tetrahedral then shape of cation formed is
(A) Tetrahedral (B) Bent (C) Linear (D) Trigonal planar



- Q.31 Shape of $\text{DCl}_4^\ominus$ is
 (A) See - Saw (B) Perfect tetrahedral
 (C) Square planar (D) none of these.

More than one may be correct

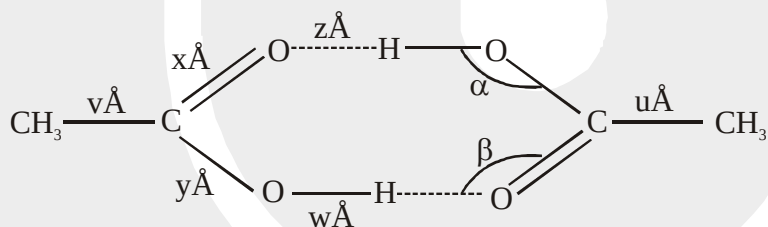
- Q.32 In the following carbon reduction process:



Note : C stands for carbon.

Which of the following option is/are **correct** regarding 'X' and 'Y'?

- (A) 'X' is neutral oxide and 'Y' is acidic oxide.
 (B) Both having $p\pi-p\pi$ bond
 (C) Both having $p\pi-p\pi$ back bond
 (D) Both having same number of π bonds.
- Q.33 In compound $\text{PX}_n\text{Y}_{5-n}$; X & Y are monovalent surrounding atoms and order of electronegativity is $X > Y$ then according to given information correct statement is / are:
 (A) If $n=3$ then bond angle between central atom and less electronegativity atom is greater than 120° .
 (B) If $n=0$, then all five bond lengths are not equal.
 (C) if $n=1$, then axial plane can have two same substituent and equatorial plane can have three same substituent.
 (D) If $n=4$ then axial plane can have three same substituent and equatorial plane can have two same substituent.
- Q.34 Consider the structure of $(\text{CH}_3\text{COOH})_2$ given below :



Select the **correct** option:

- (A) $x < y$ (B) $w < z$ (C) $\alpha < \beta$ (D) $u = v$
- Q.35 Which option is/are **correct** for perxenate ion ?
 (A) It has 6 σ bonds (B) It has 2 π bonds
 (C) It has octahedral shape (D) It is XeO_6^{4-}
- Q.36 For compound $\text{I}(\text{CF}_3)(\text{CH}_3)_2$ which statement is / are correct.
 (A) CF_3 group goes to axial position.
 (B) CH_3 group goes to equatorial position.
 (C) It has 'T' shaped geometry around I-atom.
 (D) d_{z^2} orbital is taking part in hybridisation.

- Q.37 Select the **incorrect** statements:
- (A) Hydrogen peroxy phosphate ion contains one P–O–H bond.
 (B) PH_5 exists but NH_5 does not.
 (C) $\angle \text{Cl PCl}$ and $\angle \text{Br PBr}$ bond angles are equal in PCl_3F_2 and PF_3Br_2 molecule.
 (D) According to MOT, in C_2 molecule there are two covalent bonds between two C atoms in which both are π bonds.
- Q.38 Which of the following characteristics related to the $\text{F}_2\text{C}=\text{C}=\text{CF}_2$ molecule is / are correct.
 (A) Non planar (B) Non polar
 (C) Nodal planes of π -bonds are perpendicular to each other.
 (D) Nodal planes of π -bonds are in molecular plane.
- Q.39 Select the incorrect statement for adjacent bond angle
 (A) In $:\text{CF}_2$ and OF_2 , OF_2 has higher bond angle
 (B) PF_4^+ and $\text{PF}_4^\ominus$, $\text{PF}_4^\ominus$ has higher bond angle (with respect to axial bond angle)
 (C) ClF_3 and BF_3 , ClF_3 has lower adjacent bond angle
 (D) NH_3 and PH_3 both have same bond angle

Match the column :

- Q.40 **Column I**
(Compound pair)
 (A) NH_3 and Kr
 (B) H_2O and NaCl
 (C) CH_4 and Cl_2
 (D) I–Cl and Br–F
- Column II**
(Most dominating force of interaction)
 (P) London forces
 (Q) Keesom forces
 (R) Debye forces
 (S) Ion dipole forces
- Q.41 **Column-I**
Species
 (A) $[\text{BH}_4]^-$
 (B) $[\text{BeF}_4]^{2-}$
 (C) SiC
 (D) $[\text{BF}_4]^-$
- Column-II**
Correct statement related to given species.
 (P) All atoms are p-block element
 (Q) Central atom is of s-block element
 (R) Central atom(s) is/are sp^3 hybridised
 (S) All atoms are s-block element
 (T) Central atom is of p-block element but surrounding atoms are of s-block element.
- Q.42 **Column I**
(Pair of species)
 (A) PCl_3F_2 , PCl_2F_3
 (B) BF_3 & BCl_3
 (C) CO_2 & CN_2^{-2}
 (D) C_6H_6 & $\text{B}_3\text{N}_3\text{H}_6$
- Column II**
(Identical Property in pairs of species)
 (P) Hybridisation of central atom
 (Q) Shape of molecule / ion
 (R) μ (dipole moment)
 (S) Total number of electrons
- Q.43 **Column - I**
 (A) $d_{x^2-y^2}$ atomic orbital
 (B) p_y atomic orbital
 (C) σ_{p_x} molecular orbital
 (D) $\pi_{p_y}^*$ molecular orbital
- Column - II**
 (P) One nodal plane
 (Q) Two nodal planes
 (R) $\Psi(g)$ [gerade]
 (S) $\Psi(u)$ [Ungerade]

Subjective :

- Q.44 Find the number of molecule which are planar
 (a) ClF_3 (b) SF_4 (c) XeF_6 (d) XeF_2
 (e) XeF_4 (f) H_2O (g) H_2S (h) NH_3
 (i) PH_3 (j) PCl_5 (k) PCl_3 (l) OCl_2
- Q.45 Which of the following has less than two bond order?
 NO_3^- , CO_3^{2-} , F_2 , Cl_2 , Br_2 , O_2^{2-} , O_2^- , N_2^- , O_2^{2+} , O_2^+ , Li_2^+ , He_2^+
- Q.46 Select number of molecules/ions in which central atom use its d_{z^2} orbital in hybridisation.
 XeF_6 , SOF_4 , NO_3^- , ClO_3^- , BrF_3 , IF_4^- , COCl_2 , PCl_6^-
- Q.47 Calculate the number of molecules of HF which are contained in gaseous form due to hydrogen bonding if the density of HF in gaseous phase at 1atm, 300 K is 3.248 gm/lit.
- Q.48 How many nodal plane(s) is / are present in $\pi_{2p_y}^*$ antibonding molecular orbital.
- Q.49 Find out the total number of species where the central atom is sp^2 hybridised.
 ClO_2 , NO_2 , SO_2 , ClO_3 , $\bullet\text{CF}_3$, $\bullet\text{CH}_3$, C_3O_2 , $(\text{CN})_2$, N_2H_4
- Q.50 Find out the value of z.

$$z = (a + b) - c$$
 [X = central atom]
 a = number of molecules having X-H bond
 H_3PO_3 , H_3PO_4 , H_3PO_2 , H_3BO_3 , $\text{H}_2\text{N}_2\text{O}_2$, N_2H_4
 b = number of molecules having X-X bond
 $\text{H}_2\text{S}_2\text{O}_5$, $\text{H}_2\text{S}_2\text{O}_7$, $\text{H}_2\text{S}_4\text{O}_6$
 c = number of molecules having (O-O) $^{-2}$ linkage
 $\text{H}_2\text{S}_2\text{O}_7$, H_2SO_5 , $\text{Cr}_2\text{O}_7^{-2}$, CrO_5 , $\text{H}_4\text{P}_2\text{O}_7$

ANSWER KEY

Q.1	C	Q.2	C	Q.3	A	Q.4	B	Q.5	B
Q.6	C	Q.7	B	Q.8	C	Q.9	B	Q.10	C
Q.11	A	Q.12	D	Q.13	A	Q.14	D	Q.15	C
Q.16	D	Q.17	B	Q.18	C	Q.19	D	Q.20	C
Q.21	B	Q.22	A	Q.23	D	Q.24	B	Q.25	B
Q.26	C	Q.27	A	Q.28	B	Q.29	A	Q.30	B
Q.31	B	Q.32	BD	Q.33	ABCD	Q.34	ABCD		
Q.35	ABCD	Q.36	ABCD	Q.37	ABC	Q.38	ABC	Q.39	ABD
Q.40	(A) R (B) S (C) P (D) Q			Q.41	(A) R, T ; (B) Q, R ; (C) P, R ; (D) P, R				
Q.42	(A) P, Q; (B) P, Q, R; (C) P, Q, R, S ; (D) P, Q, R, S								
Q.43	(A) Q, R, (B) P, S , (C) R , (D) Q, R					Q.44	6	Q.45	9
Q.46	5	Q.47	4	Q.48	2	Q.49	4	Q.50	0003

